

Problem - Solution Fit

Filled project document for GitHub and SkillWallet submission

Date	28 June 2026	Team ID	To be updated in SkillWallet
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction	Team Members	Devireddy Prasanna, Chokka Abitha, Navya Borugadda, Bodi Varshitha, Mulakala Harika Venkata Sridevi

Problem	Target User	Current Behavior	Proposed Solution	Fit Evidence
Flood warnings are often delayed or unclear.	Meteorologists and disaster officers	Manual review of rainfall and weather patterns.	ML classifier gives instant flood/no-flood prediction.	Supports early evacuation scenario.
Resource deployment needs prioritization.	Relief coordinators	Region-by-region judgement under pressure.	Web app supports repeated predictions for multiple areas.	Improves operational triage.
Model reliability must be validated.	Government analyst / mentor	Checks historical outcomes manually.	Notebook compares multiple algorithms and reports metrics.	XGBoost target accuracy: 96.55%.

Conclusion

The solution fits the user problem because it turns historical data and current weather readings into a quick, explainable classification that can be used for warning, resource planning, and project review.